



RoboCup@Home

Forms & Score Sheets

Registration Form



Team name: _____

Team leader name: _____

Safety first!

The speed of the robots should ensure safe operation. This means that the robot is not allowed to go faster than the local speed limits. During operation in an environment with humans it may not exceed a walking pace of humans (6 kph, 3.7 mph).

At any time when operating the robot inside and outside the scenario the owners have to stop the robot immediately if there is a remote possibility of dangerous behavior towards people and/or objects. If a referee, member of the Technical or Organizational committee, an Executive or Trustee of the federation tells the team to stop the robot, there will be no discussion and the robot has to be stopped *immediately*.

If the team does not comply, the team and its members can be excluded from the ongoing competition immediately by a decision of the RoboCup@Home Technical Committee. Furthermore, the team and its members can be banned from future competitions for a period not less than a year by a decision of the RoboCup Federation Trustee Board.

	Robot 1	Robot 2
Size of the robot		
Emergency button(s)		
Start button		
Collision avoidance		
Voice of the robot		
Robot speaker system		
Use of external devices		
ASR Bypassing method		
Other safety issues		
Custom containers.		

External Devices:

I have read and fully understood the “Safety First” statement and agree to it. Furthermore, I agree to obey the rules in the official RoboCup@Home RuleBook.

Date & time

Team leader

Score Sheet



Test: Poster Session

Team name: _____

Referee name: _____

Instructions:

1. Please watch all posters carefully.
2. There will be a team member for questions about the poster.
3. After viewing the poster and asking your questions you may have, please fill in the evaluation sheet below.
4. Is not allowed to evaluate your own team.
5. You may use the space below "Remarks:" to take notes for yourself.
6. Enter your team's name ("Team name") and your name ("referee name") on top of the sheet.
7. Sign the form using either "Referee" or "Team leader" slots at the bottom.

Evaluation sheet

Team	Overall score of poster (0-10)
Chief Scientist Office	
EPFL@Home	
eR@sers	
FBOT@Home	
Gentlebots	
Hibikino-Musashi@Home	
KIARO	
LAR@Home	
LASR	
LCASTOR	
LyonTech	
NimbRo@Home	
Pumas	
PUMAS-DSPL	
RoboCanes-VISAGE	
RoboFEI@Home	
RoBorregos rUNSWEEP	
Serious Cybernetics Corporation (SCC)	
Sinfonia Uniandes	
SKUBA	
SOBITS	
SocRob@Home	
SUTURO-VaB	
Team of Bielefeld (ToBi)	
Tech United Eindhoven	
Tidyboy-DSPL	
Tidyboy-OPL	
Tinker	
TRAIL	
UT Austin Villa@Home	

Remarks:

Date & time

Referee

Team leader

Score Sheet



Test: General Purpose Service Robot

Team name: _____

Referee name: _____

Action	Score	1 st try	2 nd try	3 rd try
Main Goal				
Understand the spoken command	3×80	_____	_____	_____
Using a custom operator	3×-20	_____	_____	_____
Request a rephrasing	6×-30	_____	_____	_____
Demonstrate a plan has been generated	3×100	_____	_____	_____
Solving the first command	250	_____	_____	_____
Solving the second command	250	_____	_____	_____
Solving the third command	250	_____	_____	_____
Solving part of a command will apply a percentage penalty according to similar scores in other tests.		_____	_____	_____
Extra Rewards				
Interleaved Task Bonus	200	_____	_____	_____
Penalties				
Bypassing speech recognition	3×-50	_____	_____	_____
Special Penalties & Bonuses				
Not attending	-500	_____	_____	_____
Using alternative start signal	-100	_____	_____	_____
Outstanding performance	149	_____	_____	_____
Recordings				
Duration (limit: 7:00 minutes)		_____	_____	_____
Restart after (sec)		_____	_____	_____
Score per try	1490	_____	_____	_____
Total Score	1639	_____	_____	_____

Remarks:

Date & time

Referee

Team leader

Score Sheet



Test: Human Robot Interaction Challenge

Team name: _____

Referee name: _____

Action	Score	1 st try	2 nd try	3 rd try
Greeting				
Detect the doorbell sound as a signal that a guest has arrived	2×30	_____	_____	_____
Open the entrance door for a guest	2×200	_____	_____	_____
Look at the person talking, when receiving a guest	2×50	_____	_____	_____
Offer a free seat to the new guest	2×100	_____	_____	_____
Look in the direction of navigation or at the navigation goal	2×15	_____	_____	_____
Tell a visual attribute of the first guest to the second guest		_____	_____	_____
Correct attribute	4×+20	_____	_____	_____
Incorrect attribute	4×-20	_____	_____	_____
Not asking non-essential questions to confirm or correct information	4×15	_____	_____	_____
Introduction				
Say name and favorite drink of each guest (during introduction)	4×30	_____	_____	_____
While introducing guests, look to the correct guest while talking about the other guest	2×50	_____	_____	_____
Guiding				
Receive the bag from the guest via handover	50	_____	_____	_____
Requesting handover assistance from the guest	-25	_____	_____	_____
Bag placed on the robot structure instead of being handed over	-50	_____	_____	_____
Following the host to the bag drop area	200	_____	_____	_____
Drop the bag while following the host	-50	_____	_____	_____
Rediscovering the operator by natural interaction	-50	_____	_____	_____
Asking the operator to wait	-50	_____	_____	_____
Guiding the robot with physical contact (take by the hand)	-150	_____	_____	_____
Drop the bag in the correct area	50	_____	_____	_____
Penalties				
Wrong guest information was memorized	4×-40	_____	_____	_____
Alternative HRI	6×-20	_____	_____	_____
Not acknowledging people	2×-200	_____	_____	_____
Special Penalties & Bonuses				
Not attending	-500	_____	_____	_____
Outstanding performance	145	_____	_____	_____
Recordings				
Duration (limit: 7:00 minutes)		_____	_____	_____
Restart after (sec)		_____	_____	_____

Score per try 1450 _____

Total Score 1595 _____

Remarks:

Date & time

Referee

Team leader

Score Sheet



Test: Pick and Place Challenge

Team name: _____

Referee name: _____

Action	Score	1 st try	2 nd try	3 rd try
Navigate to the table	15	_____	_____	_____
Correctly recognize an object	12×10	_____	_____	_____
Perceive objects on a shelf and indicate the correct placement	2×30	_____	_____	_____
Picking				
Picking up an object for transportation	12×50	_____	_____	_____
First Pick Bonus	+100	_____	_____	_____
From the floor	+30	_____	_____	_____
Cutlery	2×+50	_____	_____	_____
Plate	+100	_____	_____	_____
Dishwasher tab	+100	_____	_____	_____
Common object from auxiliary table	2×−20	_____	_____	_____
Placing				
Place an object in its designated location	12×40	_____	_____	_____
Correctly in the dishwasher	3×+70	_____	_____	_____
Next to similar objects in the cabinet	2×+20	_____	_____	_____
Area around breakfast items is not cleaned	4×−30	_____	_____	_____
In the dishwasher tab slot inside the dishwasher	+160	_____	_____	_____
Common object from auxiliary table	2×−20	_____	_____	_____
Extra Rewards				
Pull or push the dishwasher rack	2×100	_____	_____	_____
Open or close the dishwasher door without assistance	2×200	_____	_____	_____
Open milk container without assistance	400	_____	_____	_____
Pour cereal or milk into the bowl without assistance	2×200	_____	_____	_____
Penalties				
Objects thrown or dropped while placing	12×−40	_____	_____	_____
Breakfast not served in a typical meal setting	−50	_____	_____	_____
Objects dropped on the floor	12×−40	_____	_____	_____
Spilling cereal and milk while pouring	2×−100	_____	_____	_____
Human assistance: object repositioned by a person	12×−30	_____	_____	_____
Human assistance: handover	24×−100	_____	_____	_____
Human assistance: environment changes (per item)	−40	_____	_____	_____
Human assistance: opening milk container	−0	_____	_____	_____
Human assistance: moving dishwasher door or rack	−0	_____	_____	_____
Special Penalties & Bonuses				
Not attending	−500	_____	_____	_____
Using alternative start signal	−100	_____	_____	_____
Outstanding performance	351	_____	_____	_____
Recordings				
Duration (limit: 7:00 minutes)		_____	_____	_____
Restart after (sec)		_____	_____	_____
Score per try	3515	_____	_____	_____
Total Score	3866	_____		
Remarks:				

Date & time

Referee

Team leader

Score Sheet



Test: Restaurant

Team name: _____

Referee name: _____

Action	Score	1 st try	2 nd try	3 rd try
Regular Rewards				
Detect calling or waving customer	2×80	_____	_____	_____
Reach a customer's table	2×80	_____	_____	_____
Human Assistance: Being guided to a table	2×-80	_____	_____	_____
Understand and confirm the order received to the customer	2×160	_____	_____	_____
Alternative HRI	2×-80	_____	_____	_____
Not making eye-contact when taking the order	2×-60	_____	_____	_____
Communicate the order to the barman	2×80	_____	_____	_____
Picking up the requested items from the <i>Kitchen-bar</i>	4×100	_____	_____	_____
First Pick Bonus	+100	_____	_____	_____
Human assistance: Asking the Barman to handover object to the robot	4×-100	_____	_____	_____
Return to the customer table with the order	2×80	_____	_____	_____
Serve the order to the customer	4×100	_____	_____	_____
First Place Bonus	+100	_____	_____	_____
Human assistance: Guest needing to take the object from a tray or the robot's hand	4×-100	_____	_____	_____
Extra Rewards				
Use an unattached tray to transport	2×200	_____	_____	_____
Penalties				
Not reaching the bar (barman has to move from behind the bar to interact with the robot)	2×-60	_____	_____	_____
Human Assistance: Asking for directional confirmation	2×-30	_____	_____	_____
Human Assistance: Being told/pointed where a table/ <i>Kitchen-bar</i> is located	2×-40	_____	_____	_____
Special Penalties & Bonuses				
Not attending	-500	_____	_____	_____
Using alternative start signal	-100	_____	_____	_____
Outstanding performance	236	_____	_____	_____
Recordings				
Duration (limit: 15:00 minutes)		_____	_____	_____
Restart after (sec)		_____	_____	_____
Score per try	2360	_____	_____	_____
Total Score	2596	_____	_____	_____

Remarks:

Date & time

Referee

Team leader

Score Sheet



Test: Doing Laundry

Team name: _____

Referee name: _____

Action	Score	1 st try	2 nd try	3 rd try
Main Goal				
Navigating to the laundry area	15	_____	_____	_____
Picking up a piece of clothing from the basket	100	_____	_____	_____
Picking up multiple at once	-100	_____	_____	_____
Folding a piece of clothing	800	_____	_____	_____
Quality penalties	-800	_____	_____	_____
Human assistance: Flattening/Arranging clothing before folding	-200	_____	_____	_____
Human assistance during folding will apply a percentage penalty according to the amount of help	-800	_____	_____	_____
Extra Rewards				
Opening the washing machine door	300	_____	_____	_____
Removing one or more pieces of clothing from the washing machine	300	_____	_____	_____
The clothing touches the floor	-200	_____	_____	_____
Using the basket for transportation	300	_____	_____	_____
Laundry is dropped or otherwise lost during transportation.	-200	_____	_____	_____
Folding additional clothes (per item)	5×400	_____	_____	_____
Quality penalties	-400	_____	_____	_____
Human assistance: Flattening/Arranging clothing before folding	5×-100	_____	_____	_____
Human assistance during folding will apply a percentage penalty according to the amount of help	5×-400	_____	_____	_____
Stacking folded clothes neatly (per item)	6×100	_____	_____	_____
Penalties				
Human assistance: environment changes (per item)	-40	_____	_____	_____
Special Penalties & Bonuses				
Not attending	-500	_____	_____	_____
Using alternative start signal	-100	_____	_____	_____
Outstanding performance	441	_____	_____	_____
Recordings				
Duration (limit: 7:00 minutes)		_____	_____	_____
Restart after (sec)		_____	_____	_____

Score per try 4415 _____

Total Score 4856 _____

Remarks:

Date & time

Referee

Team leader

Score Sheet



Test: Final Demonstration

Team name: _____

Referee name: _____

Action	Score	Single try
Main Goal (can be repeated unlimited times)		
Find and clearly state an encountered problem	3× 150	_____
Find repeated problem category	−100	_____
Human Assistance: Asking for location of a problem	−150	_____
Solve a problem	10× 650	_____
Solving repeated problem category for the 2nd time	−300	_____
Solving repeated problem category for the 3rd (or more) time	−500	_____
Instructing a human to perform parts of the task will apply a percentage penalty according to similar scores in other tests.	−650	_____
Solving only parts of a GPSR command will apply a percentage penalty according to similar scores in other tests.		_____
Moving the laundry basket	600	_____
Opening the Door of the Apartment	600	_____
Closing the Dishwasher	300	_____
Custom:		
	00	_____
Custom:		
	00	_____
Custom:		
	00	_____
Penalties		
Restart (only applies if the robot continues scoring afterwards)	−50	_____
Recordings		
Duration (limit: 10:00 minutes)		_____
Restart after (sec)		_____

Total Score 8450 _____

Referee Information: Note each problem (category, item, location) and mark if they are stated by the robot in remarks

Remarks:

Date & time

Referee

Team leader